Mohammad Abbas Kazmi

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WORK EXPERIENCE

ADOBE New York, NY

Software Engineer *August 2024 – Present*

- Built an agentic search service and Adobe's first context-aware recommendation app by having agents select search facets from document and parent-app context, driving adoption across 380 customer accounts.
- Engineered an agent tracing pipeline capturing 100K+ monthly traces via LangFuse, then fine-tuned agent LLMs on the labeled data, boosting task success rate from 25% to 70% and cutting prompt tuning from weeks to hours.
- Developed an agent evaluation system with guardrails against unstructured outputs and silent failures, increasing result relevance by 45% and cutting undetected failures by 75%.
- Built and adapted agentic features for 10+ enterprise customers across pharma, retail, travel, and tech, tailoring existing agent capabilities to each customer's requirements while owning setup and escalation support directly.
- Designed an automated security alerting system that detects and strips prompt injection and malicious content before it reaches the LLM, cutting triage time 60% and catching 20K+ risks monthly.
- Architected a reusable agent orchestration harness in CrewAI and LangGraph, standardizing cost tracking and observability across teams and cutting agent setup time from days to minutes.
- Created AI-enhanced UI features in TypeScript and ReactJS, including recent-search components and suggested-content panels for agentic experiences.
- Implemented internal and public-facing APIs in Java and FastAPI to support AI-powered features and scalable services.
- Deployed cloud-native infrastructure using Azure Functions, Blob Storage, and AWS Bedrock to run production AI workloads.

Software Engineer Intern, Machine Learning *June 2023 – September 2023*

- Spearheaded development of machine learning models to predict user checkouts by leveraging high frequency click stream data.
- Led end to end pipeline development on Databricks by utilizing SQL and Python to transform raw data into actionable insights.
- Engineered robust and efficient data processing solutions using Apache Spark and Airflow to streamline data ingestion, preprocessing, and orchestration.
- Achieved high accuracy models (~93%) through various machine learning methods and integrated via meticulous software engineering.
- Strategically planned model deployments and microservice integrations to enable real-time predictions on Adobe.com.

Software Engineer Intern, eCommerce Experiences *June 2022 – September 2022*

- Developed a Testing Framework for Adobe Checkout Team developers to test or validate any new, existing, or error scenarios using mock data.
- Restructured the Checkout application's distributed system design by incorporating AWS S3, EC2 to store/return mock responses.
- Designed a Testing API in TypeScript with Apollo GraphQL that integrates within existing APIs to create a seamless testing experience.
- Implemented Redis Caching through an ioredis server to allow for quick retrieval of repeated calls and log/review analytics.

EDUCATION

University of California, Davis Davis, CA

Bachelor of Science in Computer Science and Engineering

Concentrations: Deep learning and AI

Relevant Coursework: Deep Learning, Computer Vision, Operating Systems, Computer Architecture, Algorithms & Design

TECHNICAL SKILLS

Programming Languages: Python, Java, TypeScript, C++, Go

OS & Tools/Frameworks: Linux, Docker, SQL, MongoDB Atlas, GraphQL, Apache Spark, React.js, AWS, Azure

AI/Agent Frameworks: LangChain, LangGraph, CrewAI, LangFuse, PyTorch, AWS Bedrock, Azure OpenAI